

WASP-10b

R-1961

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_190625 · created 2026-08-06T22:56:31+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458659.837 +/- 0.0043 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.117 +/- 0.062
Transit depth (Rp/Rs)^2	1.37 +/- 1.45 [%]
Orbital Inclination (inc)	86.44 +/- 1.93
Airmass coefficient 1 (a1)	1.29 +/- 0.87
Airmass coefficient 2 (a2)	-0.117 +/- 0.094
Scatter in the residuals of the lightcurve fit is	65.75 %
Best Comparison Star	None
Optimal Aperture	3.8
Optimal Annulus	8.86
Transit Duration (day)	0.061 +/- 0.03

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.117 ± 0.062 vs 0.1592 (deviation 0.68σ)
Duration fit vs published	0.061 d vs 0.0946 d (deviation 1.12σ)
Mid-time O–C	-8.2 min
Depth SNR	1.9
Residual scatter	65.75 %

Light curve

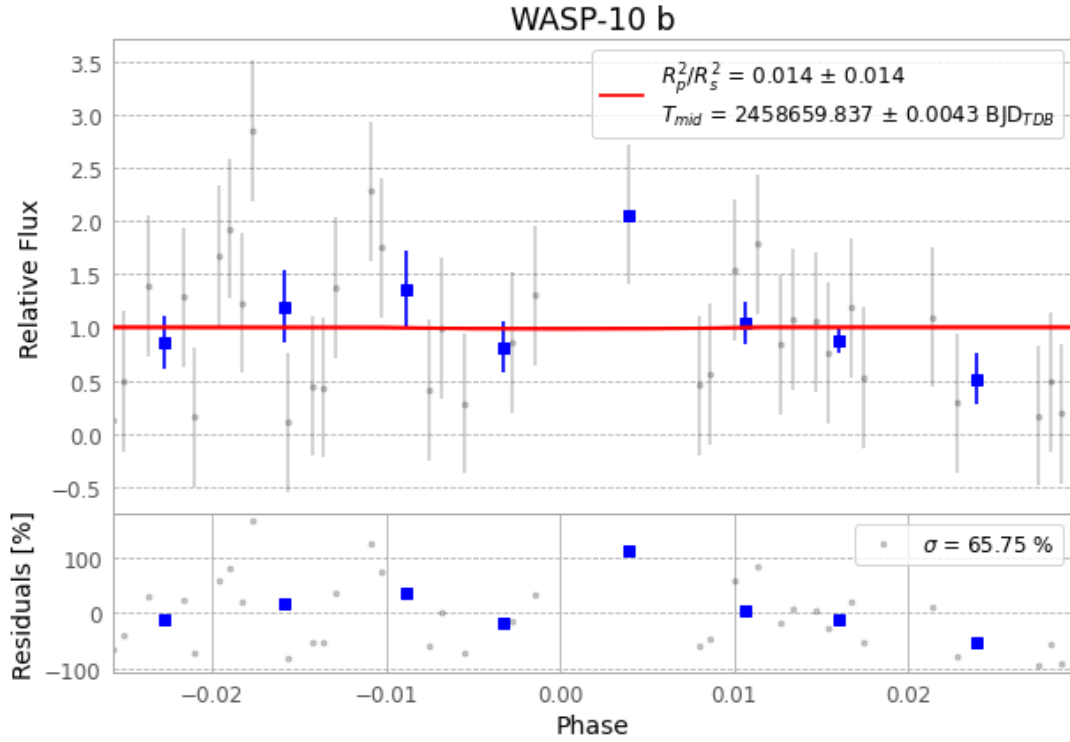


Figure 1 — FinalLightCurve_WASP-10 b_2019-06-24.png (SHA-256 b7ab1cb00c9ad1c2...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [565, 82], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 39ed0d15b479e22d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

83 FITS frames (55 MB), WASP-10190625060912.FITS → WASP-10190625101512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-190625.log (58f3455938d7...), FinalLightCurve_WASP-10 b_2019-06-24.png (b7ab1cb00c9a...), FinalLightCurve_WASP-10 b_2019-06-24.csv (2b3e3fc6f055...)

Compute resources

Wall time 117.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-06 22:56Z.